

CS5182 Mid-term Quiz 2025-26 B

19:00 – 21:00

Student ID:	
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This paper contains 5 pages (including this page).

Answer all questions.

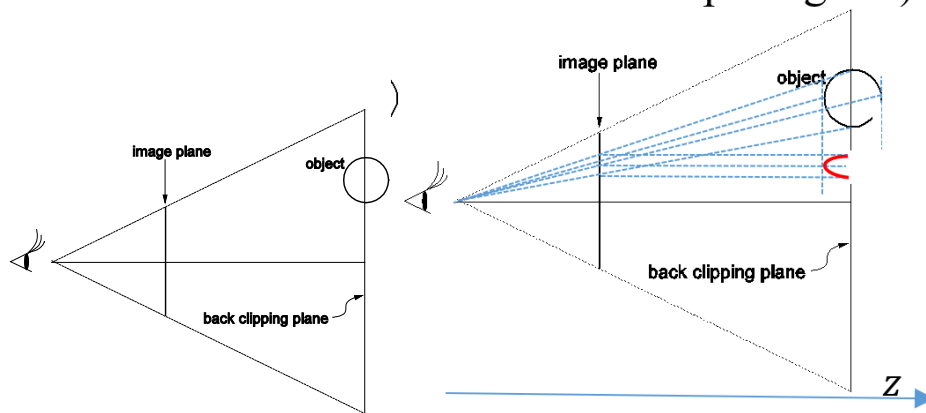
This is a **closed-book** exam.

[illegible]

1. Most standard algorithms, like Sutherland-Hodgman, are designed for clipping polygons against a rectangular "view window". Propose a conceptual approach for clipping a 3D polygon mesh against a **spherical** clip region. [8 points]

Answer: approximate the spherical clip region using polyhedron. Clip the 3D polygon against each polygon face of the polyhedron. The two endpoints of an edge are distributed at two sides of a typical polygon face of the polyhedron, and there will be an intersection point.

2. Show the output of the following object after perspective **transformation** and clipping. (Indicate how you obtain the output figure.) [10 points]



Select some points on the circle and then transform the points. Connect the transformed points according to the order to approximate the transformed circle.

3. A 3D object is rotated by 90 degrees about an axis passing from $(-1, 2, 1)$ to $(1, 2, 1)$. It is then non-uniformly scaled $(2, 3, 4)$ times relative to the origin along x, y, z dimensions. Write in their proper order the individual matrices composing this transformation. [12 points]

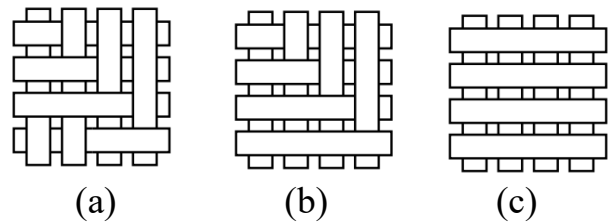
$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4. You are tasked with modeling a high-fidelity 3D human face for a real-time application (like a video game) versus a high-precision industrial part for a manufacturing simulation. Compare the suitability of Polygon Meshes versus Parametric Surfaces for these two scenarios. Which representation would you choose for each, and how would the choice impact the "Rasterization" stage of the rendering pipeline? [8 points]

Answer: For the Human Face, polygon meshes are preferred because real-time hardware (GPUs) is specifically optimized for rasterizing triangles. For the Industrial Part, parametric surfaces are superior because they provide exact mathematical precision and can be scaled to any resolution without "faceting" (jagged edges).

Parametric surfaces cannot be directly rasterized by standard pipelines. They must first undergo a Tessellation stage where the smooth surface is converted into a high-density polygon mesh before entering the Rasterization stage

5. Which of the following scenes in Figure 3 would cause problems for the painter's algorithm? Please justify your answer and suggest a method to address the issue. (These drawing are in image space; each rectangle is a single primitive) [6 points]



Answer: (a) there is cyclic overlapping. We can use z-buffering algorithm.

6. If you move the Center of Projection (COP) further away from the projection plane while keeping the object in the same position, how does this affect the "Foreshortening" effect and the distance between vanishing points? How would this change the "depth sensation" of the viewer? [9 points]

Answer: Foreshortening: As the COP moves further away, the relative difference in distance from the camera between the front and back of the object decreases. This reduces foreshortening, making the object appear "flatter."

Vanishing Points: The vanishing points move further apart toward infinity.

Depth Sensation: The "depth sensation" is reduced; the image begins to resemble a parallel (orthographic) projection where size does not diminish with distance.

7. A graphics programmer renders a high-gloss plastic sphere using a low-resolution polygon mesh. They apply the Phong Illumination Model at each vertex and then use Gouraud Shading to fill the triangles. They notice that as they move the camera, the shiny highlight "flickers" or disappears when it should be in the center of a triangle. [12 points]

(a) Explain why Gouraud shading fails to represent the specular highlight correctly in this scenario, even though the Phong illumination model is being used.

Answer: Gouraud shading calculates the full Phong illumination (including the specular component) only at the vertices and then linearly interpolates those resulting colors across the triangle. Because a specular highlight is a non-linear effect, it often falls in the middle of a triangle face. If the highlight does not hit a vertex, the interpolation will miss it entirely, or if it hits one vertex, it will be "smeared" unrealistically toward the others.

(b) Propose a solution using Phong Shading (normal interpolation) and explain why this technique resolves the "flickering" highlight without increasing the number of polygons in the mesh.

Answer: Instead of interpolating colors, Phong Shading interpolates the surface normal vectors across the triangle. By having a unique, interpolated normal at every pixel, the GPU can re-calculate the Phong Illumination Model for every single pixel on the screen. This allows the non-linear specular highlight to be accurately captured and rendered even if it falls far from any vertex, providing a smooth and stable highlight on low-polygon models.

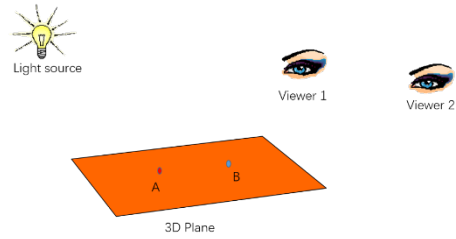
8. The standard rendering pipeline typically performs Back-face removal before Clipping and Hidden Surface Removal. Explain the computational advantage of this specific ordering. Would the final image be different if "Back-face removal" were performed after "Rasterization"? [10 points]

Answer: Advantage: Performing back-face removal early acts as a "high-level cull." Since roughly 50% of an opaque object's polygons face away from the camera, removing them early saves the pipeline from performing expensive clipping and rasterization calculations on geometry that is guaranteed to be invisible.

If performed after rasterization, the final image would remain the same (because the Z-buffer would eventually discard those fragments), but it would be highly inefficient.

9. As shown in the right figure,

(a) Assume the 3D plane has the uniform diffuse reflection property. If the light source is a directional light source, what is relationship between the diffuse reflections at points A and B? Please justify your answer [4 points]



Let I_d^A and I_d^B be the diffuse reflections at points A and B, respectively. Let θ_A and θ_B be the incident angles of the light rays at points A and B, i.e., the angle between the light ray and the surface normal vector, respectively.

As it is a directional light source, the incoming light rays are parallel, i.e., $\theta_A = \theta_B$. Based on the formula of computing diffusion reflection, we can know $I_d^A = I_d^B$.

(b) Assume the 3D plane has the uniform diffuse reflection property. If the light source is a point light source, what is relationship between the diffuse reflections at points A and B? Please justify your answer [4 points]

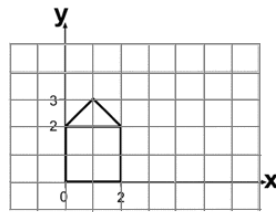
As it is a point light source, the incoming light rays are divergent, i.e., the incident angles of the light rays at points A and B are different. If your answer is $I_d^A > I_d^B$, your justification should be $\theta_A < \theta_B$; if your answer is $I_d^A < I_d^B$, your justification should be $\theta_A > \theta_B$.

(c) Assume the 3D plane is a glossy surface. If the light source is a point light source, what is the relationship between the specular reflections at point B observed by viewers 1 and 2? Please justify your answer. [7 points]

Let I_s^1 and I_s^2 be the specular reflections at viewpoints 1 and 2, respectively. Let θ_1 be the angle between the ideal reflected ray and viewing direction towards Viewer 1, and θ_2 the angle between the ideal reflected ray and viewing direction towards Viewer 2. If your answer is $I_s^1 > I_s^2$, your justification is $\theta_1 < \theta_2$. If your answer is $I_s^1 < I_s^2$, your justification is $\theta_1 > \theta_2$.

10. The following figure shows a 2D house model and a 2D affine transformation matrix $\mathbf{M}$. Sketch the transformed house and label the coordinates of all vertices.

[10 points]



$$\mathbf{M} = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Observe that $\mathbf{M} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

So this transformation first rotates the house 270 degree around the origin and then moves it 2 units along the x-axis and 1 unit along the y-axis.

--End--